$$H_2SO_4 \longrightarrow H_2O + SO_2 + \underset{\text{nascent oxygen}}{(O)}$$

Sulphuric acid oxidises elements such as carbon, sulphur and phosphorus. It also oxides bromide and iodide to bromine and iodine respectively.

$$C + 2H_2SO_4 \longrightarrow 2SO_2 + 2H_2O + CO_2$$
$$S + 2H_2SO_4 \longrightarrow 3SO_2 + 2H_2O$$
$$P_4 + 10H_2SO_4 \longrightarrow 4H_3PO_4 + 10SO_2 + 4H_2O$$
$$H_2S + H_2SO_4 \longrightarrow SO_2 + 2H_2O + S$$
$$H_2SO_4 + 2HI \longrightarrow 2SO_2 + 2H_2O + I_2$$
$$H_2SO_4 + 2HBr \longrightarrow 2SO_2 + 2H_2O + Br_2$$

Reaction with metals: Sulphuric acid reacts with metals and gives different product depending on the reactants and reacting condition.

Dilute sulphuric acid reacts with metals like tin, aluminium, zinc to give corresponding sulphates.

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2 \uparrow$$
$$2Al + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2 \uparrow$$

Hot concentrated sulphuric acid reacts with copper and lead to give the respective sulphates as shown below.

$$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + 2H_2O + SO_2 \uparrow$$
$$Pb + 2H_2SO_4 \longrightarrow PbSO_4 + 2H_2O + SO_2 \uparrow$$

Sulphuric acid doesn't react with noble metals like gold, silver and platinum.

Reaction with salts: It reacts with different metal salts to give metal sulphates and bisulphates.

$$KCl + H_2SO_4 \longrightarrow KHSO_4 + HCl$$
$$KNO_3 + H_2SO_4 \longrightarrow KHSO_4 + HNO_3$$
$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$
$$2NaBr + 3H_2SO_4 \longrightarrow 2NaHSO_4 + 2H_2O + Br_2 + SO_2$$

Reaction with organic compounds: It reacts organic compounds such as benzene to give sulphonic acids.

$$\underset{\text{Benzene}}{C_6H_6} + H_2SO_4 \longrightarrow \underset{\text{Benzene sulphonic acid}}{C_6H_5SO_3H} + H_2O$$

Uses of sulphuric acid:

1. Sulphuric acid is used in the manufacture of fertilisers, ammonium sulphate and super phosphates and other chemicals such as hydrochloric acid, nitric acid etc...
2. It is used as a drying agent and also used in the preparation of pigments, explosives etc..

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